#3.9

$$A = \begin{pmatrix} 5 & -4 & -2 \\ 8 & -5 & -4 \\ -4 & 3 & 3 \end{pmatrix}$$

Solve for eigenvalues

$$\det(A-\lambda I) = (5-\lambda)\begin{vmatrix} -5-\lambda & -4 \\ 3 & 3-\lambda \end{vmatrix} + 3\begin{vmatrix} 8 & -4 \\ -4 & 3-\lambda \end{vmatrix} - 2\begin{vmatrix} 8 & -5-\lambda \\ -4 & 3 \end{vmatrix}$$

$$= (5-\lambda)\left[(-5-\lambda)(3-\lambda)+12\right] + 3\left[8(3-\lambda)-16\right] - 2\left[24-4(5+\lambda)\right]$$

$$= (\lambda-1)^3 = 0$$

Thus, $\lambda_1 = \lambda_2 = \lambda_3 = 1$

Solving for eigenvectors:

$$(A-\lambda I)\vec{x} = \vec{0}$$

$$\begin{pmatrix} 4 & -3 & -2 \\ 8 & -6 & -4 \\ -4 & 3 & 2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$R_2 = R_2 - 2R_1$, $R_3 = R_3 + R_1$

$$\begin{pmatrix} 4 & -3 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \vec{0}$$

$$\Rightarrow 4x_1 - 3x_2 - 2x_3 = 0$$

$$\Rightarrow x_1 = \tfrac{3}{4}x_2 + \tfrac{1}{2}x_3$$

$$\vec{x} = x_2\begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix} + x_3\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

$AM > GM$, thus matrix isn't diagonalizable.

$$(A-I)(A-I) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \text{ thus}$$

$$e^{tA} = e^t\left(I + t(A-I)\right)$$

$$= e^t\begin{pmatrix} 4t+1 & -3t & -2t \\ 8t & 1-6t & -4t \\ -4t & 3t & 2t+1 \end{pmatrix}$$

#3.11

$$A = \begin{pmatrix} 4 & -1 & 2 \\ 1 & 2 & 2 \\ -2 & 2 & 1 \end{pmatrix}$$

Solving for eigenvalues

$$\det(A-\lambda I) = (4-\lambda)\left[(2-\lambda)(1-\lambda)-4\right] + 5-\lambda + 2\left[2+2(2-\lambda)\right]$$

$$= \lambda^3 - 7\lambda^2 + 15\lambda - 9$$

$$= (1-\lambda)(\lambda-3)^2$$

$\lambda_1 = 1$

$\lambda_2 = \lambda_3 = 3$

For λ_1:

$$(A-I)\vec{v}_1 = \vec{0}$$

$$\begin{pmatrix} 3 & -1 & 2 \\ 1 & 1 & 2 \\ -2 & 2 & 0 \end{pmatrix}\vec{v}_1 = 0$$

$$\Rightarrow \vec{v}_1 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

For λ_2:

$$(A-3I)\vec{v}_2 = \vec{0}$$

$$\begin{pmatrix} 1 & -1 & 2 \\ 1 & -1 & 2 \\ -2 & 2 & -2 \end{pmatrix}\vec{v}_2 = \vec{0}$$

$$\Rightarrow \vec{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$AM = GM$, thus A isn't diagonalizable.

$$(A-3I)\vec{v}_3 = \vec{v}_2 \quad \Rightarrow \vec{v}_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

We use Jordan blocks

$$J = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{pmatrix} \Rightarrow e^{Jt} = \begin{pmatrix} e^t & 0 & 0 \\ 0 & e^{3t} & te^{3t} \\ 0 & 0 & e^{3t} \end{pmatrix}$$

$$P = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} = \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

$$P^{-1} = \left(\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ -1 & 0 & 1 & 0 & 0 & 1 \end{array}\right)$$

$R_2 := R_1$

$\Rightarrow$

$R_3 := R_1$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{array}\right)$$

$R_1 := R_2$

$\Rightarrow$

$R_1 := R_3$

Swap R_2 & R_3

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & -1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 & 0 \end{array}\right)$$

$$Pe^{Jt} = \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}\begin{pmatrix} e^t & 0 & 0 \\ 0 & e^{3t} & te^{3t} \\ 0 & 0 & e^{3t} \end{pmatrix}$$

$$(Pe^{Jt})P^{-1} = \begin{pmatrix} e^t & e^{3t} & e^{3t}(t-1) \\ e^t & e^{3t} & te^{3t} \\ -e^t & 0 & e^{3t} \end{pmatrix}\begin{pmatrix} -1 & 1 & -1 \\ 1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}$$

$$Pe^{Jt}P^{-1} = \begin{pmatrix} e^{3t}(2-t)-e^t & e^{3t}(t-1)+e^t & e^{3t}-e^t \\ e^{3t}(1-t)-e^t & te^{3t}+e^t & e^{3t}-e^t \\ e^t-e^{3t} & e^{3t}-e^t & e^t \end{pmatrix}$$

#4.2

Let $A=\begin{pmatrix}5&3\\3&5\end{pmatrix}$, then we can find the solution to $\vec{x}'(t)=A\vec{x}(t)+e^{-2t}\begin{pmatrix}t\\t\end{pmatrix}$ w/ $\vec{x}(0)=\begin{pmatrix}0\\1\end{pmatrix}$ by duhamel's principle. First, we solve e^{tA}

$$\det(A-\lambda I)=(5-\lambda)(5-\lambda)-9$$
$$=\lambda^2-10\lambda+16$$
$$=(\lambda-8)(\lambda-2)$$

Thus, we have eigenvalues $\lambda_1=1, \lambda_2=8$

- For $\lambda_1=2$:

$$(A-2I)\vec{v}_1=\begin{pmatrix}3&3\\3&3\end{pmatrix}\vec{v}_1=\vec{0}$$
$$\Rightarrow \vec{v}_1=\begin{pmatrix}-1\\1\end{pmatrix}$$

- For $\lambda_2=8$:

$$(A-8I)\vec{v}_2=\begin{pmatrix}-3&3\\3&-3\end{pmatrix}\vec{v}_2=\vec{0}$$
$$\Rightarrow \vec{v}_2=\begin{pmatrix}1\\1\end{pmatrix}$$

Thus, we have

$$P=\begin{pmatrix}-1&1\\1&1\end{pmatrix} \quad D=\begin{pmatrix}2&0\\0&8\end{pmatrix}$$
$$P^{-1}=\frac{1}{2}\begin{pmatrix}-1&1\\1&1\end{pmatrix} \quad e^{Dt}=\begin{pmatrix}e^{2t}&0\\0&e^{8t}\end{pmatrix}$$

and,

$$e^{tA}=Pe^{Dt}P^{-1}$$
$$=\frac{1}{2}\begin{pmatrix}-e^{2t}&e^{8t}\\e^{2t}&e^{8t}\end{pmatrix}\begin{pmatrix}-1&1\\1&1\end{pmatrix}$$
$$=\frac{1}{2}\begin{pmatrix}e^{2t}+e^{8t}&e^{8t}-e^{2t}\\e^{8t}-e^{2t}&e^{2t}+e^{8t}\end{pmatrix}$$

Thus, we have

$$e^{tA}\vec{x}_0=\frac{1}{2}\begin{pmatrix}e^{8t}-e^{2t}\\e^{8t}+e^{2t}\end{pmatrix}$$

Notice that the forcing term $\vec{g}(t)=te^{-2t}\vec{v}_2$.

Hence, we have

$$\int_0^t e^{(t-s)A}\vec{g}(s)\,ds=\int_0^t e^{8(t-s)}se^{-8s}\vec{v}_2\,ds$$
$$=e^{8t}\vec{v}_2\int_0^t se^{-16s}\,ds$$
$$=e^{8t}\vec{v}_2\left[-\frac{s}{16}e^{-16s}+\frac{1}{16}\int e^{-16s}\,ds\right]$$
$$=e^{8t}\vec{v}_2\left[-\frac{s}{16}e^{-16s}-\frac{1}{256}e^{-16s}\right]_0^t$$
$$=e^{8t}\vec{v}_2\left(-\frac{t}{16}e^{-16t}-\frac{1}{256}e^{-16t}+\frac{1}{256}\right)$$
$$=\left(-\frac{t}{16}e^{-8t}-\frac{1}{256}e^{-8t}+\frac{1}{256}e^{8t}\right)\vec{v}_2$$

Therefore, our final general solution is

$$\vec{x}(t)=\frac{1}{2}\begin{pmatrix}e^{8t}-e^{2t}\\e^{8t}+e^{2t}\end{pmatrix}+\left(-\frac{1}{16}te^{-8t}-\frac{1}{256}e^{-8t}+\frac{1}{256}e^{8t}\right)\begin{pmatrix}1\\1\end{pmatrix}$$

4.8a

Let $\vec{v}(x,y) = \begin{pmatrix} (x+y)(x-y-1) \\ (x+y-2)(x-y+1) \end{pmatrix} = \begin{bmatrix} x^2 - x - y^2 - y \\ x^2 - x - y^2 + 3y - 2 \end{bmatrix}$

Set $\vec{v}(x,y) = \vec{0}$ to find equilibrium points

$(x+y)(x-y-1) = 0$

Case A ($x = -y$):

$(x+y-2)(x-y+1) = -2(-2y+1)$

$= 4y - 2 = 0$

$y = \frac{1}{2}$

$\vec{x}_1 = \frac{1}{2}\begin{pmatrix} -1 \\ 1 \end{pmatrix}$

Case B ($x = y+1$):

$(x+y-2)(x-y+1) = (2y-1)(2)$

$4y = 2$

$y = \frac{1}{2}$

$\Rightarrow \vec{x}_2 = \frac{1}{2}\begin{pmatrix} 3 \\ 1 \end{pmatrix}$

Now, we compute the Jacobian

$D_v(x,y) = \begin{bmatrix} 2x-1 & -2y-1 \\ 2x-1 & -2y+3 \end{bmatrix}$

Evaluate at $\vec{x}_1$

$D_v(-\frac{1}{2}, \frac{1}{2}) = \begin{pmatrix} -2 & -2 \\ -2 & 2 \end{pmatrix}$

Computing the eigenvalues

$\det(D_v(-\frac{1}{2}, \frac{1}{2}) - \lambda I) = (-2-\lambda)(2-\lambda) - 4$

$0 = \lambda^2 - 8$

$\lambda = \pm\sqrt{8}$

Eigenvalues have opposite signs, thus this equilibrium point is unstable.

Evaluate at $\vec{x}_2$:

$D_v(\frac{3}{2}, \frac{1}{2}) = \begin{pmatrix} 2 & -2 \\ 2 & 2 \end{pmatrix}$

Computing the eigenvalues

$\det(D_v(\frac{3}{2}, \frac{1}{2}) - \lambda I) = (2-\lambda)(2-\lambda) + 4$

$0 = \lambda^2 - 4\lambda + 8$

using quadratic formula:

$\frac{4 \pm \sqrt{16 - 4(8)(1)}}{2(1)} = \frac{4 \pm \sqrt{-16}}{2} = \frac{4 \pm 4i}{2} = 2 \pm 2i$

Thus, the eigenvalues are $\lambda = 2 \pm 2i$,

The real portion is positive, thus this equilibrium point is also unstable.

Let $\vec{v}(x,y) = \begin{pmatrix} (x+y-2)(x-y+1) \\ (x+y)(x-y-1) \end{pmatrix} = \begin{pmatrix} x^2+3y-x-y^2-2 \\ x^2-x-y^2-y \end{pmatrix}$

We solve $\vec{v}(x,y) = \vec{0}$ for equilibrium pts

Case A $(x = 2-y)$:

$(x+y)(x-y-1) = (2)(1-2y)$

$0 = 2-4y$

$y = \frac{1}{2} \Rightarrow x = 2 - \frac{1}{2} = \frac{3}{2}$

$\Rightarrow \vec{x}_1 = \frac{1}{2}\begin{pmatrix} 3 \\ 1 \end{pmatrix}$

Case B: $(x = y-1)$

$(x+y)(x-y-1) = (2y-1)(-2)$

$0 = -4y+2$

$y = \frac{1}{2} \Rightarrow x = \frac{1}{2} - 1 = -\frac{1}{2}$

$\vec{x}_2 = \frac{1}{2}\begin{pmatrix} -1 \\ 1 \end{pmatrix}$

Compute the Jacobian

$D_v(x,y) = \begin{pmatrix} 2x-1 & 2y+3 \\ 2x-1 & -2y-1 \end{pmatrix}$

At $\vec{x}_1$:

$D_v\left(\frac{3}{2}, \frac{1}{2}\right) = \begin{pmatrix} 2 & 2 \\ 2 & -2 \end{pmatrix}$

$\det\left(D_v\left(\frac{3}{2}, \frac{1}{2}\right) - \lambda I\right) = (2-\lambda)(-2-\lambda) - 4$

$= \lambda^2 - 8$

$\lambda = \pm\sqrt{8}$

We again have opposite signed eigenvalues, thus $\vec{x} = \left(\frac{3}{2}, \frac{1}{2}\right)$ is unstable.

At $\vec{x}_2$:

$D_v\left(-\frac{1}{2}, \frac{1}{2}\right) = \begin{pmatrix} -2 & 2 \\ -2 & -2 \end{pmatrix}$

$\det\left(D_v\left(-\frac{1}{2}, \frac{1}{2}\right) - \lambda I\right) = (-2-\lambda)(-2-\lambda) + 4$

$= \lambda^2 + 4\lambda + 8$

Applying quadratic formula

$\lambda = \frac{-4 \pm \sqrt{16 - (4)(1)(8)}}{2(1)} = \frac{-4 \pm \sqrt{-16}}{2} = -2 \pm 2i$

The real part of the eigenvalues is strictly negative, thus, $\left(-\frac{1}{2}, \frac{1}{2}\right)$ is asymptotically stable.

#4.9a

Let $\vec{v}(x,y) = \begin{pmatrix} x(1-y) \\ y(1+2x) \end{pmatrix}$

Solve $\vec{v}(x,y) = \vec{0}$ for equilibrium pts

Case A ($x=0$):

$y(1+2x) = y = 0$

$\Rightarrow \vec{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Case B ($y=1$):

$y(1+2x) = 1+2x$

$x = -\frac{1}{2}$

$\Rightarrow \vec{x}_2 = \begin{pmatrix} -1/2 \\ 1 \end{pmatrix}$

Compute the Jacobian

$D_v(x,y) = \begin{pmatrix} 1-y & -x \\ 2y & 1+2x \end{pmatrix}$

At $\vec{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$:

$D_v(0,0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$

$\lambda_1 = \lambda_2 = 1$

Both eigenvalues are positive, thus, $\vec{x}_1 = (0,0)$ is unstable.

At $\vec{x}_2 = \begin{pmatrix} -1/2 \\ 1 \end{pmatrix}$:

$D_v(-\frac{1}{2}, 1) = \begin{pmatrix} 0 & 1/2 \\ 2 & 0 \end{pmatrix}$

$\det(D_v(-\frac{1}{2}, 1) - \lambda I) = \lambda^2 - 1$

$\lambda = \pm 1$

One eigenvalue is positive, so the pt $\vec{x}_2 = (-\frac{1}{2}, 1)$ is also unstable.

#4.9b

Let $\vec{v}(x,y) = \begin{pmatrix} y+2xy \\ x-xy \end{pmatrix}$

Swapping the components x and y, we are left with the same equilibrium pts $\vec{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and $\vec{x}_2 = \begin{pmatrix} -1/2 \\ 1 \end{pmatrix}$

Compute the Jacobian

$$D_v(x,y) = \begin{pmatrix} 2y & 2x+1 \\ 1-y & -x \end{pmatrix}$$

At $\vec{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$:

$$D_v(0,0) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\det(D_v(0,0)) = \lambda^2 - 1$$

$$\lambda_1 = 1, \lambda_2 = -1$$

The eigenvalue λ_1 is positive, thus the pt $\vec{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ is unstable.

At $\vec{x}_2 = \begin{pmatrix} -1/2 \\ 1 \end{pmatrix}$:

$$D_v\left(-\tfrac{1}{2}, 1\right) = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$$

$$\lambda_1 = 2, \lambda_2 = \tfrac{1}{2}$$

Both eigenvalues are positive, thus, the pt $\vec{x}_2 = \left(-\tfrac{1}{2}, 1\right)$ is also unstable.